



COM 682

Cloud Native Development

*Decomposing a Cloud: Virtualization, Containerization
and Serverless Execution*

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On-Prem

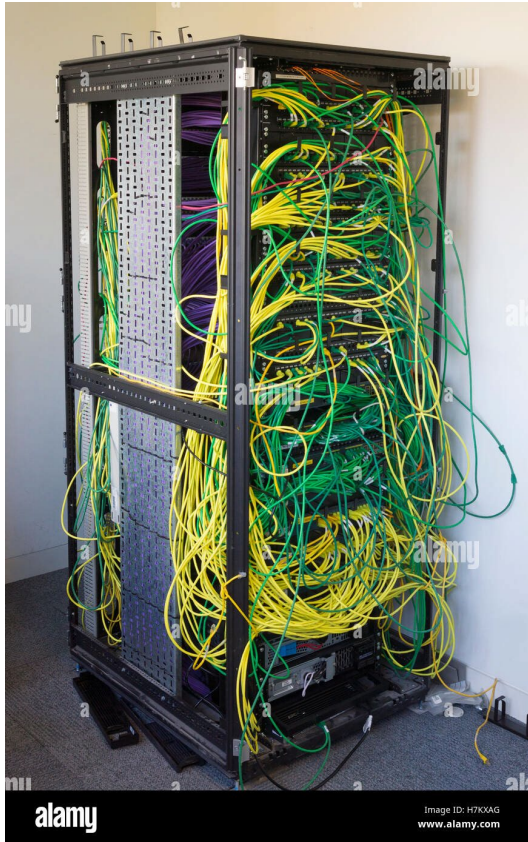
VS



Cloud



On Premises Characteristics



Monolithic, centralized

Design for predictable scalability

Relational database

Strong consistency

Serial and synchronized processing

Design to avoid failures (MTBF)

Occasional big updates

Manual management

Snowflake servers

Modern Cloud Characteristics



Decomposed, de-centralized

Design for elastic scale

Polyglot persistence (mix of storage technologies)

Eventual consistency

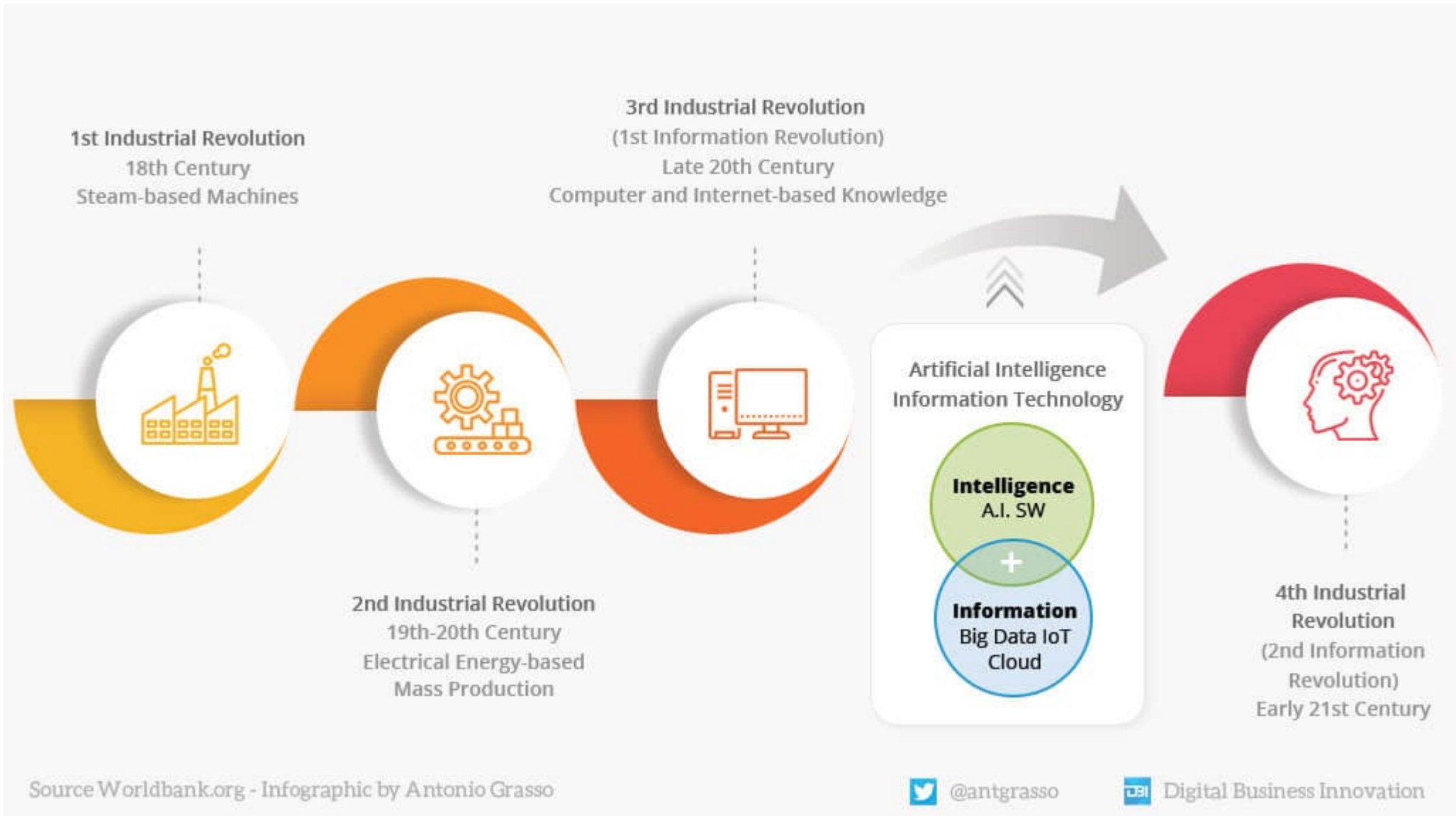
Parallel and asynchronous processing

Design to avoid failure (MTTR)

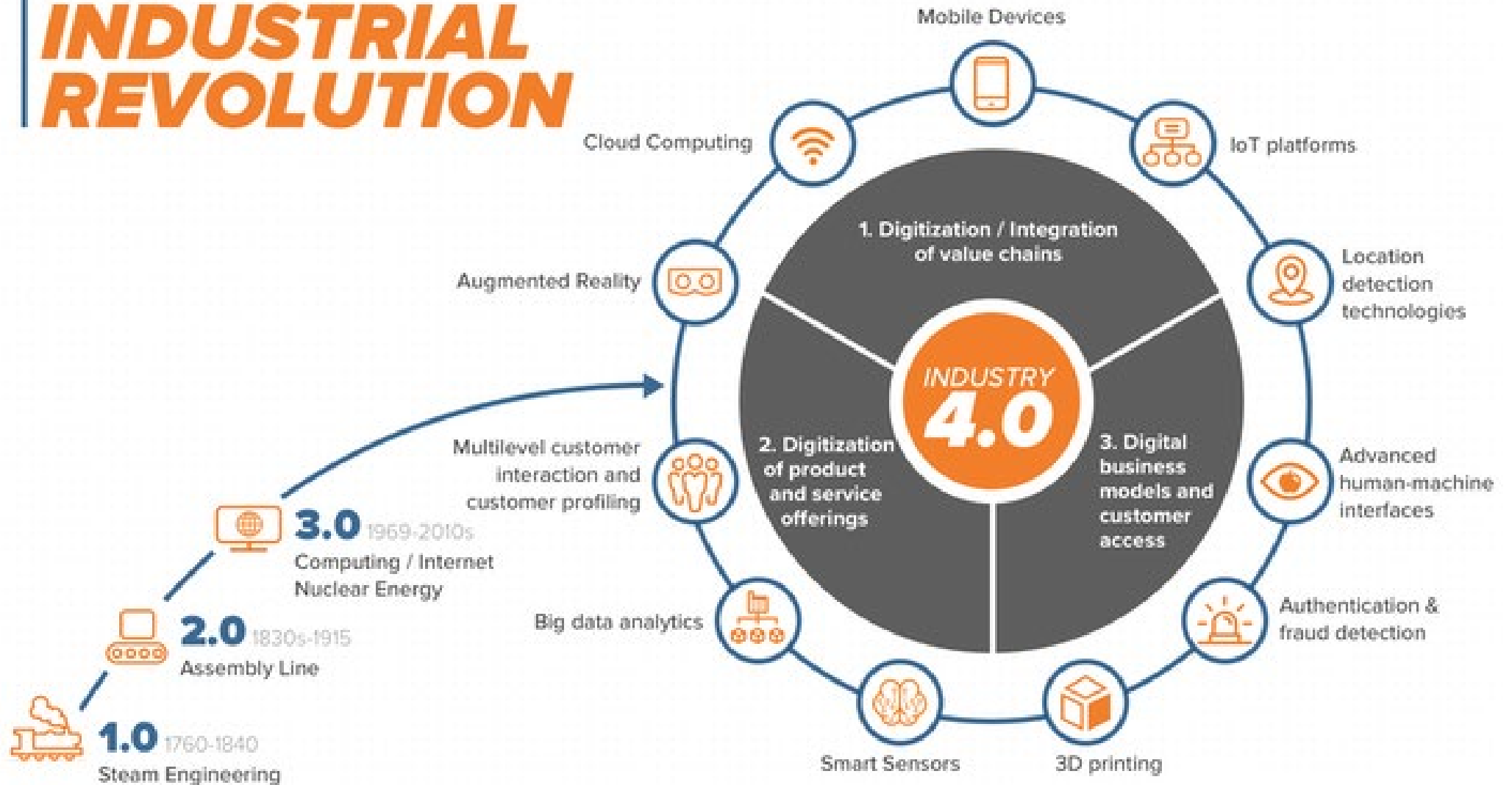
Frequent small updates

Automated self-management

Immutable infrastructure



FOURTH INDUSTRIAL REVOLUTION



Private, Public and Hybrid clouds

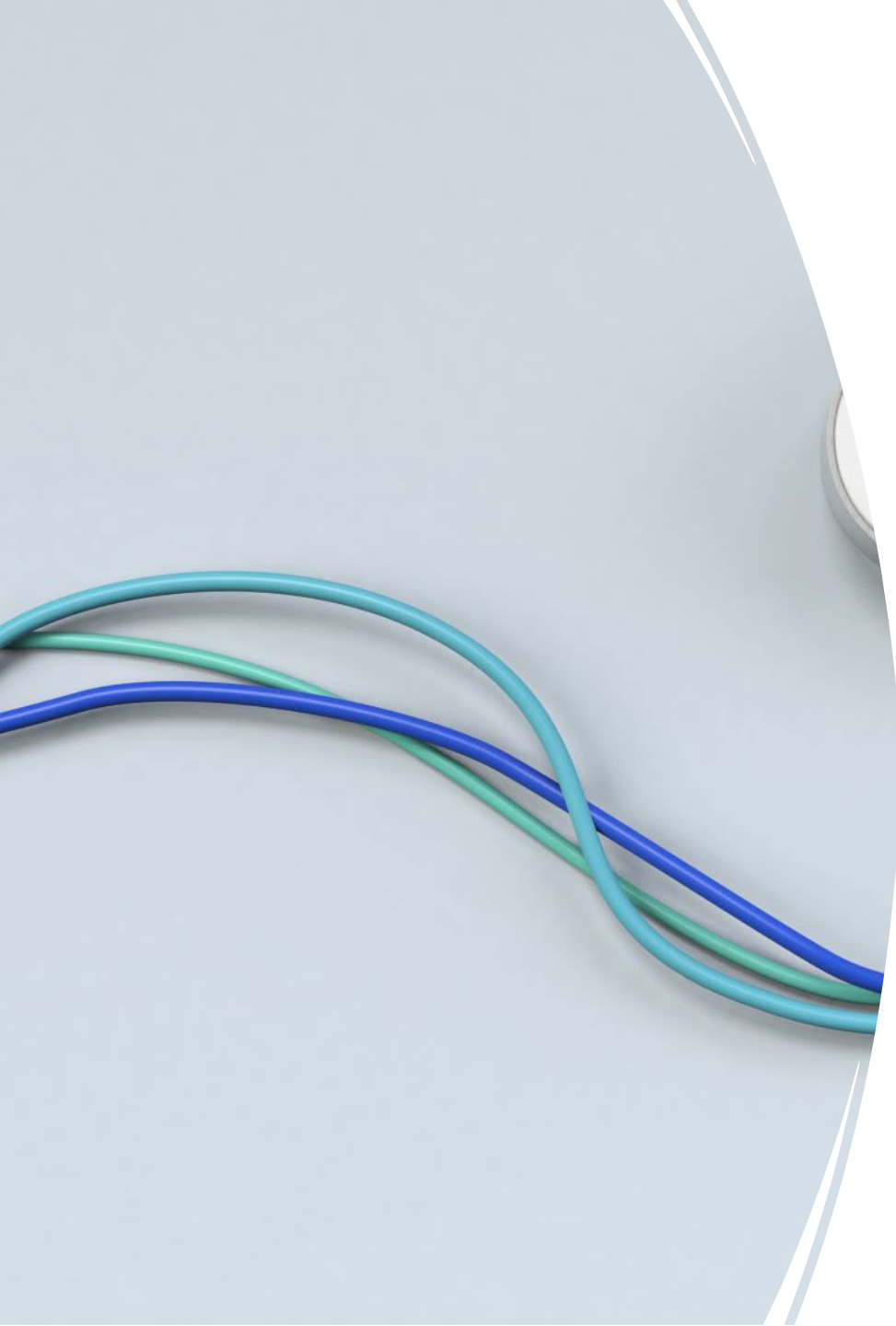
There are three well-known deployment models for cloud computing: public, private, and hybrid clouds.

A **public cloud** is owned by a cloud provider but is made available to the public.

A **private cloud** is typically owned by an organization, which also controls the access to the cloud.

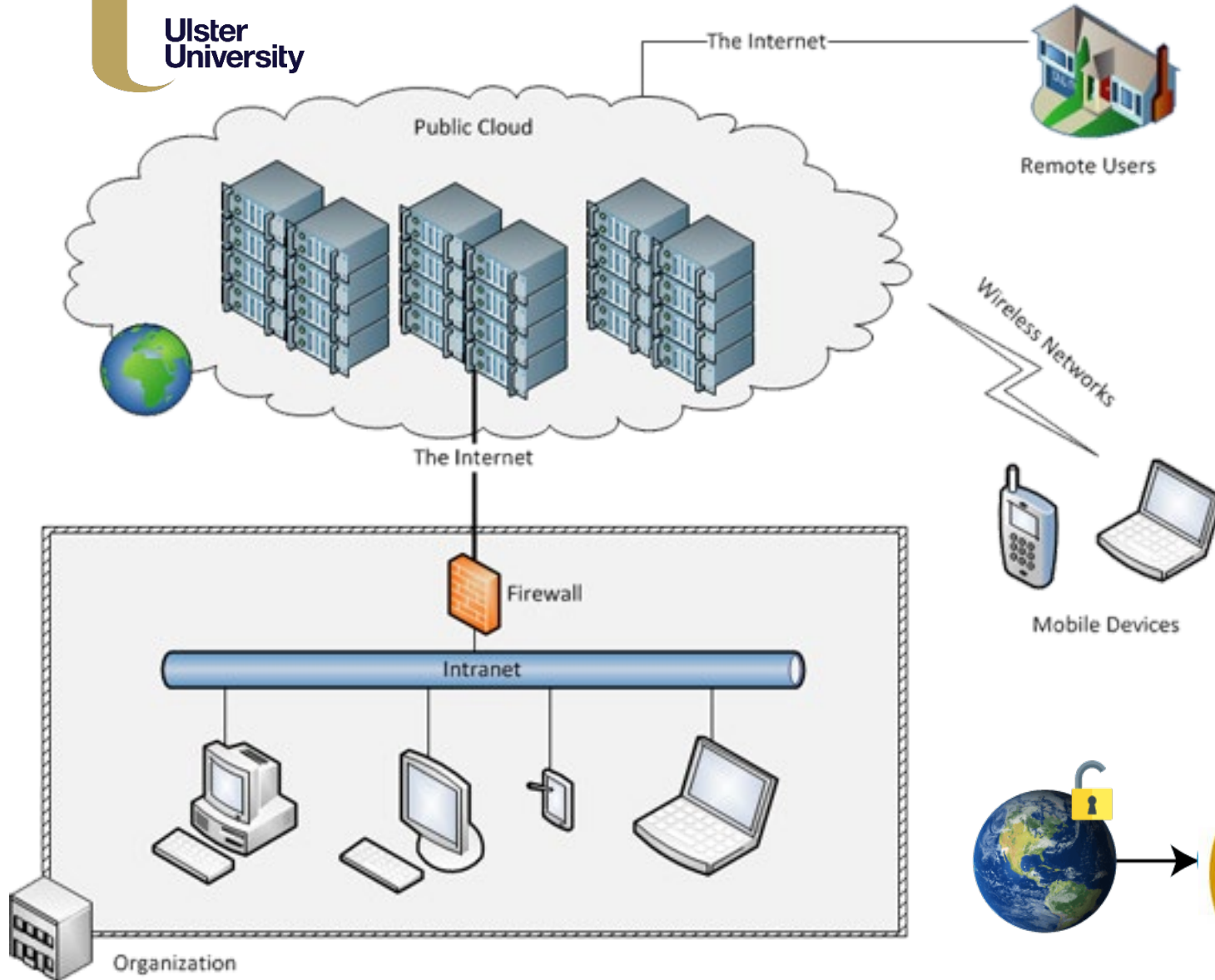
A **hybrid cloud** is a combination of public and private clouds.

We will discuss the different types in terms of **ownership, infrastructure, end-user availability, cost, security, and data location.**



Public Cloud

- In a public cloud, the cloud infrastructure is owned by a cloud provider and is accessible to the public over the internet.
- The **cloud provider hosts the cloud infrastructure**, and end users can access it remotely without the need to buy and set up a working environment (such as buying hardware and software).
- Public cloud **resources are shared** among different end users.
- Public cloud users are **typically charged for the duration for which these services are used**. However, public cloud charge models vary across providers.
- The **security and terms of use are defined by the provider**, and hence, end users must work within the constraints of the provider when using its services.

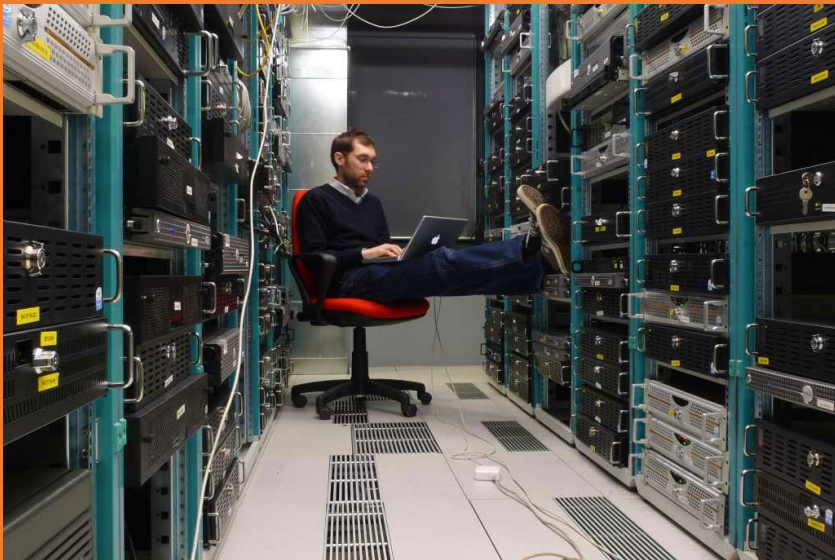


Public Cloud

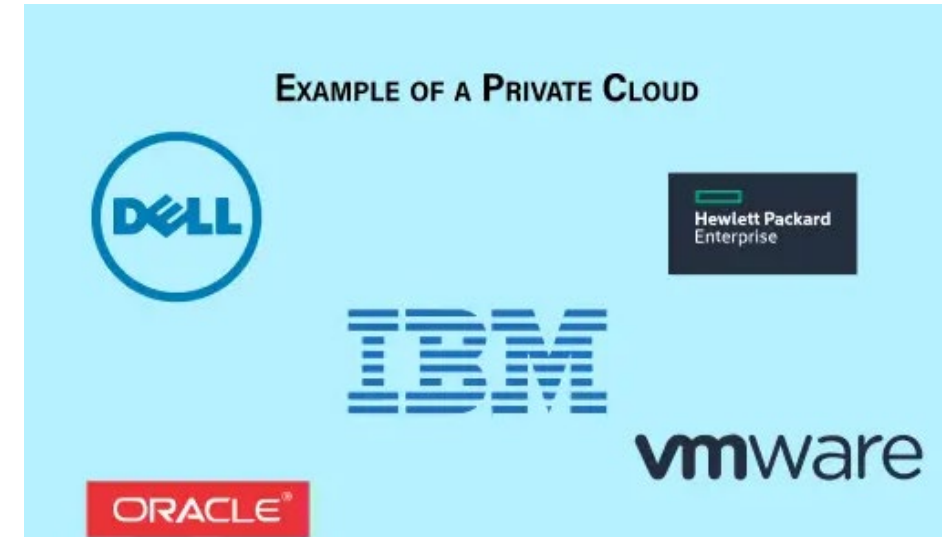
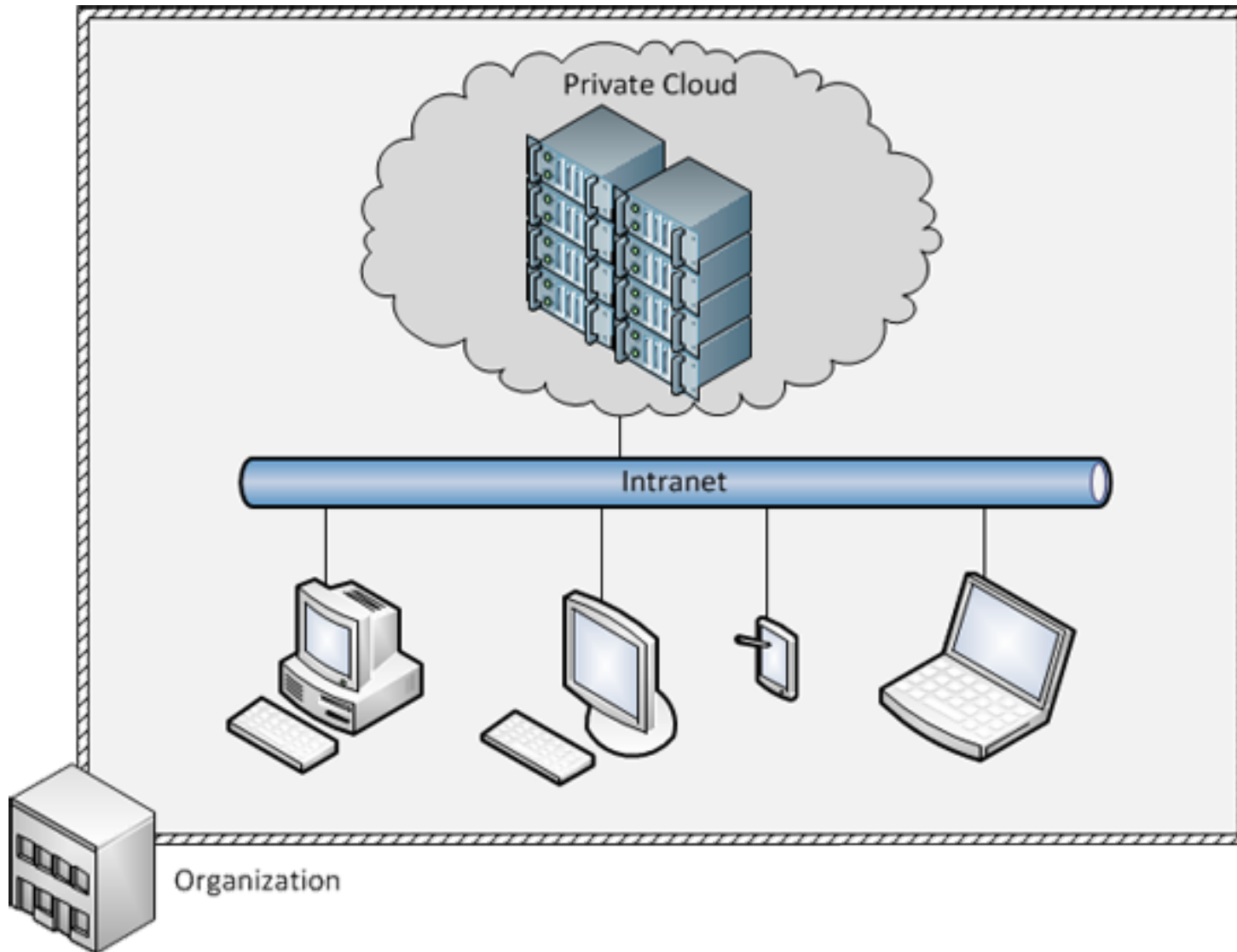


Private Cloud

- In this second type of cloud, the cloud infrastructure is **owned by an organization**.
- The infrastructure is **accessible to specific users** via the organization's intranet.
- The **cloud environment needs to be procured, set up, operated, and maintained by the organization** itself. The private cloud resources are typically shared among an organization's end users.
- Unlike the public cloud, **security and terms of use** for a private cloud are defined by the organization.
- The **entire infrastructure is located in the organization**, and hence, security can be compliant with the organization's policies.



Private Cloud



Azure Stack Hub

Bringing the agility and innovation of cloud computing to your on-premises environment.

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What is Azure Stack Hub?

Part of the Azure Stack portfolio, Azure Stack Hub broadens Azure to let you run apps in an on-premises environment and deliver Azure services in your datacenter. As organizations race to digitally transform, many are finding they can move faster by using public cloud services to build on modern architectures and refresh legacy apps. Many workloads, however, must remain on-premises—for example, due to technological and regulatory obstacles. Microsoft has you covered with broad hybrid cloud options and cloud innovation for all your workloads, wherever they reside.



Your own autonomous cloud that you can run completely or partially disconnected from the internet and the public cloud



Ability to use consistent tools, experiences, and app models and easily transfer them between Azure and Azure Stack Hub



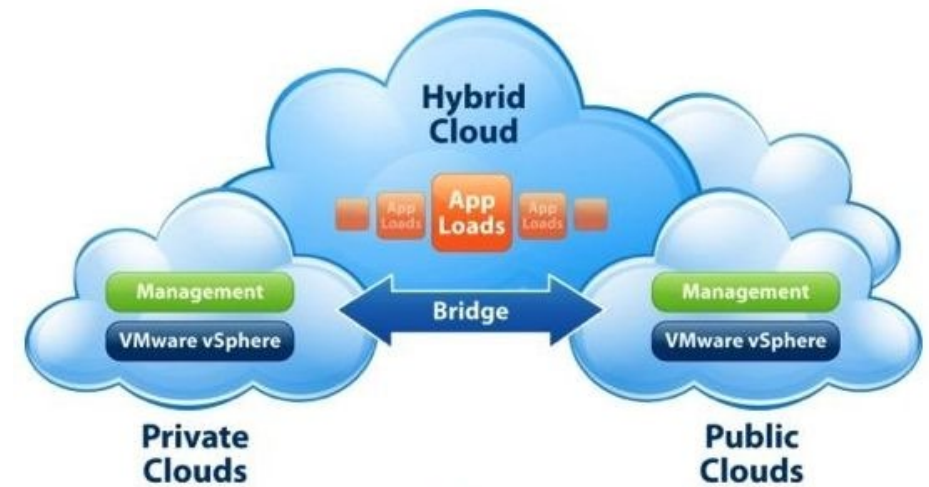
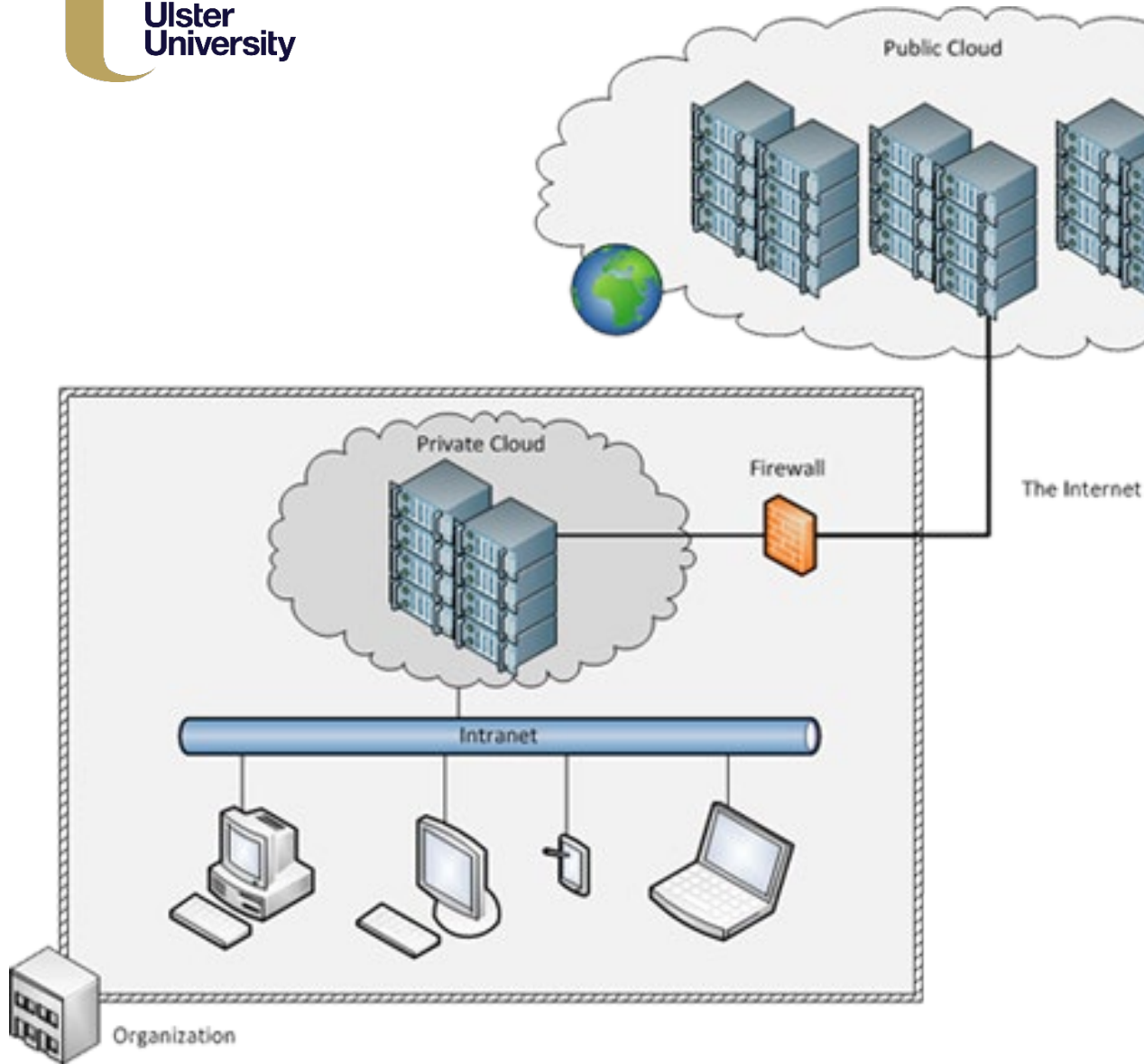
One Azure ecosystem giving you integrated systems from Hewlett Packard Enterprise, Dell EMC, Cisco, Lenovo, and others



Hybrid Cloud

- In a hybrid cloud, the infrastructure includes an **owned private cloud and a leased public cloud**.
- Hybrid clouds enable the idea of "**cloud bursting**," in which an organization uses its private cloud for most of its needs and dynamically provisions resources in the public cloud when utilization exceeds the capacity of its private cloud.

Hybrid Cloud



Hybrid Cloud

Here are some good examples of hybrid cloud usage:

1. Netflix: Netflix uses a hybrid cloud to deliver its streaming services. It leverages both public cloud for scalability and private cloud for sensitive data storage and content delivery optimization [2].

2. Digital Transformation: Many organizations adopt hybrid cloud for digital transformation initiatives. They utilize public clouds for agility and private clouds for data security and compliance requirements [5].

3. Development and Testing: Hybrid cloud enables organizations to create a flexible environment for development and testing. Public cloud resources are used during development, while the private cloud provides a secure test environment [5]



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AWS Outposts Family

Run AWS infrastructure and services on premises for a truly consistent hybrid experience

[Get started with Outposts](#)

Model Comparison

Public cloud

- No capital expenditures to scale up.
- Applications can be quickly provisioned and deprovisioned.
- Organizations pay only for what they use.

Private cloud

- Hardware must be purchased for start-up and maintenance.
- Organizations have complete control over resources and security.
- Organizations are responsible for hardware maintenance and updates.

Hybrid cloud

- Provides the most flexibility.
- Organizations determine where to run their applications.
- Organizations control security, compliance, or legal requirements

"As you move up the stack from IaaS to PaaS to Serverless, Azure manages more infrastructure, but you must still secure your applications and data."

As you consider and evaluate public cloud services, it's critical to understand the shared responsibility model and which security tasks the cloud provider handles and which tasks you handle.

The workload responsibilities vary depending on whether the workload is hosted on Software as a Service (SaaS), Platform as a Service (PaaS), Infrastructure as a Service (IaaS), or in an on-premises datacenter.

Responsibility		SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Customer	Shared	Customer	Customer
	Network controls	Customer	Shared	Customer	Customer
	Operating system	Customer	Customer	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

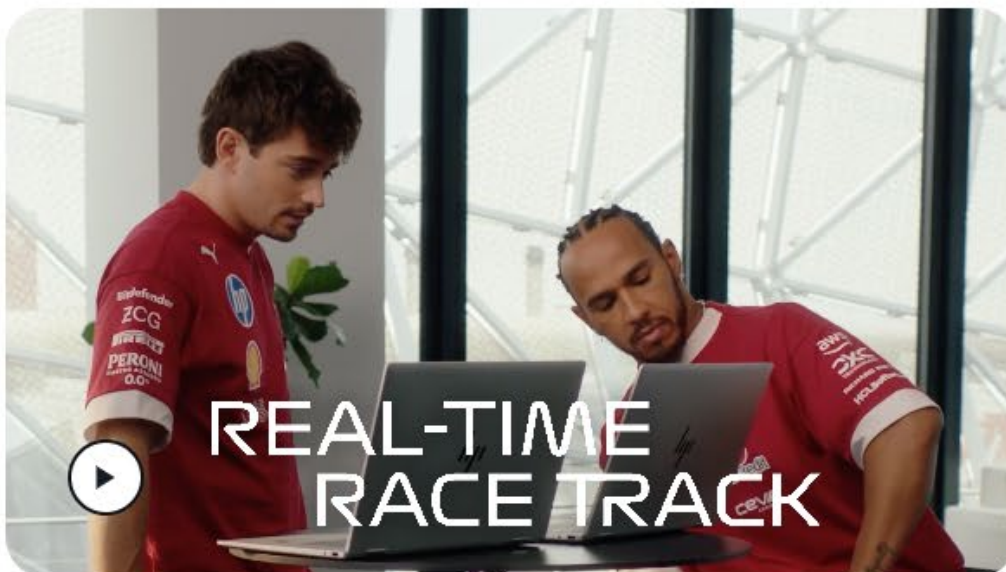
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F1 on AWS

Explore new, unique, and innovative ways for fans to enjoy F1 through the power of machine learning, artificial intelligence and cloud technologies.



Global Partner of FORMULA 1



AWS x Formula 1®: Made for This

Watch as Scuderia Ferrari HP driver Charles Leclerc and seven-time world champion Lewis Hamilton design their own race tracks with AWS generative AI. See these legendary drivers discuss their favorite turns and explore how AI can be applied to motorsport.

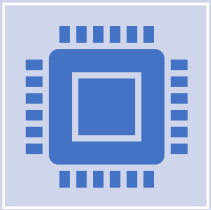
Logical/Physical division - Cloud services and implementation

In a cloud context, the main approaches to subdividing, abstracting and managing these underlying resources are :

- A. Virtual machines - IaaS**
- B. Containers - IaaS**
- C. Distributed App Services - PaaS**
- D. Functions - PaaS**

Virtual Machines



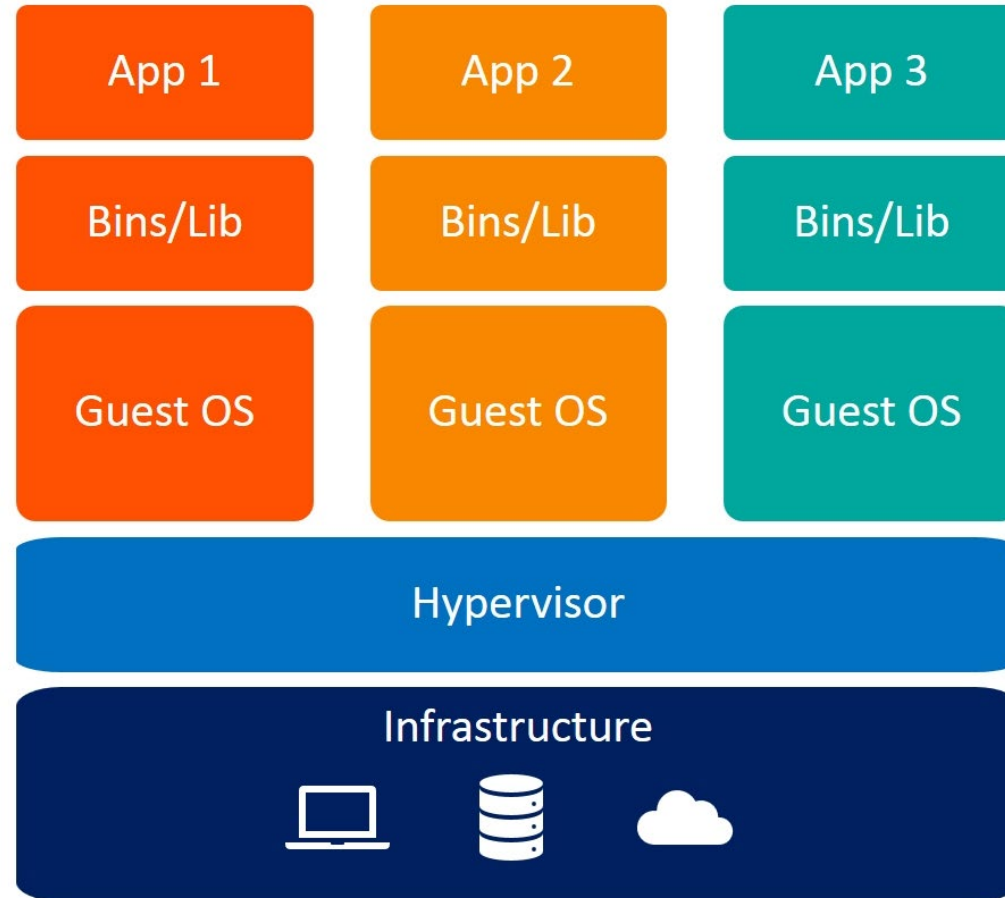


Virtual machines are software emulations of physical computers. They include a **virtual processor, memory, storage, and networking resources**. VMs host an **operating system**, and you can **install and run software just like a physical computer**.

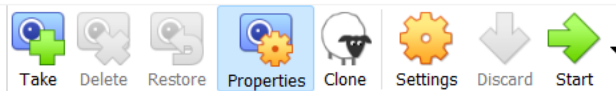


With Azure Virtual Machines, you can create and use VMs in the cloud. Virtual Machines provides infrastructure as a service (IaaS) and can be used in different ways. **When you need total control over an operating system and environment, VMs are an ideal choice.**

Virtual Machines



Virtual Machines



64

Bee-Box
 Powered Off

Name Taken

Current State

Attributes

Information

Name:

Enter a name for the new snapshot...

Description:

Take

Reset

Use Cases for Azure Virtual Machines

Examples of when to use VMs:

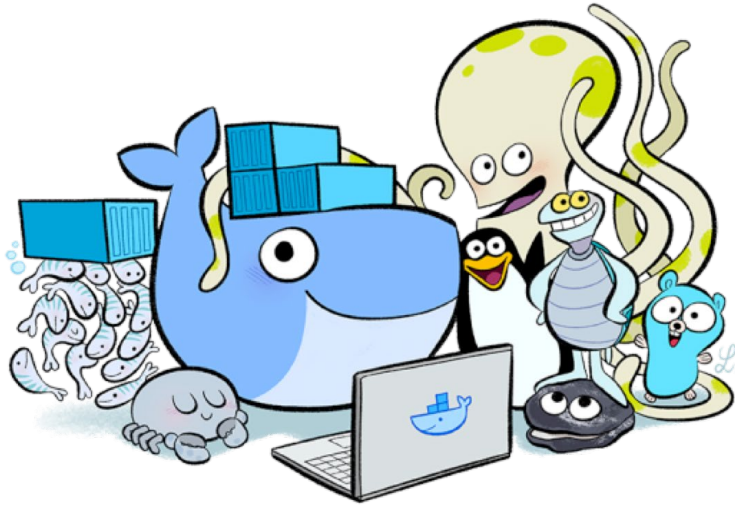
- **During testing and development.** VMs provide a quick and easy way to create different OS and application configurations. Test and development personnel can then easily delete the VMs when they no longer need them.
- **When running applications in the cloud.** The ability to run certain applications in the public cloud as opposed to creating a traditional infrastructure to run them can provide substantial economic benefits. For example, **an application might need to handle fluctuations in demand.** Shutting down VMs when you don't need them or quickly starting them up to meet a sudden increase in demand means you pay only for the resources you use.

Use Cases for Azure Virtual Machines

Examples of when to use VMs:

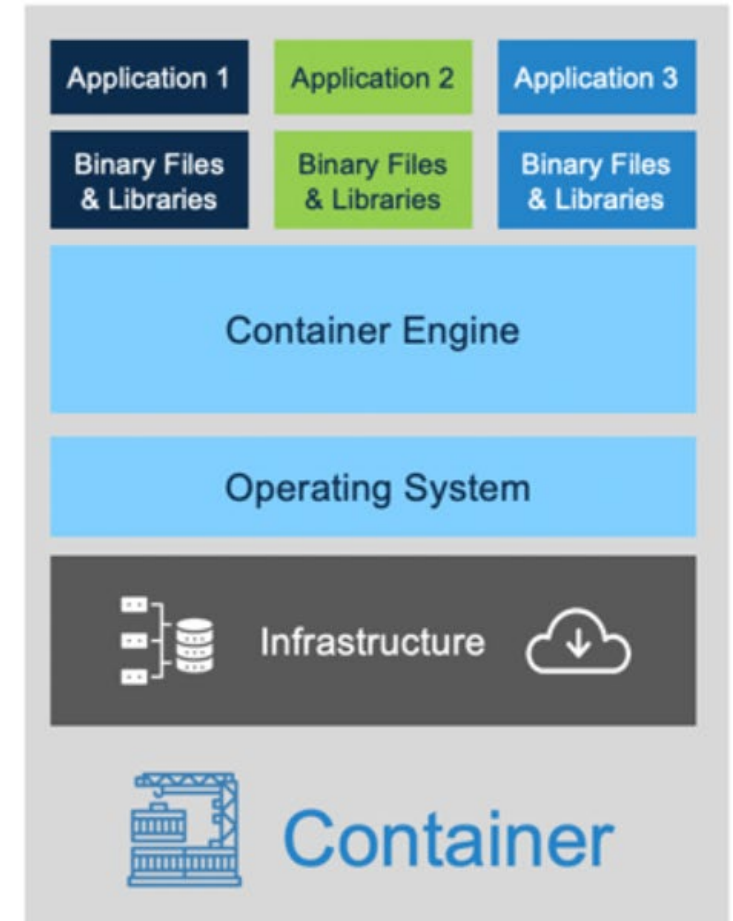
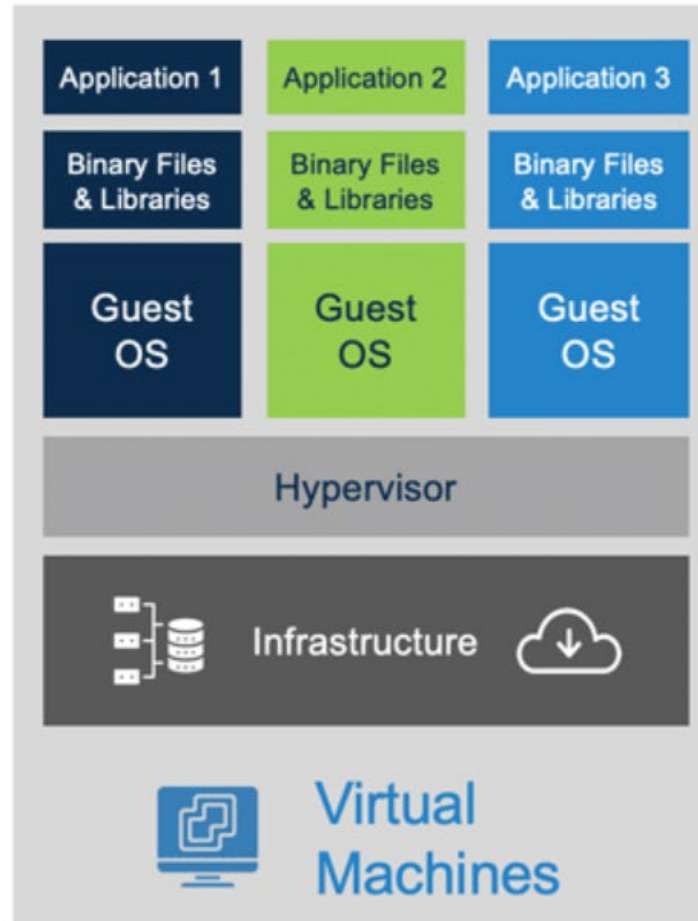
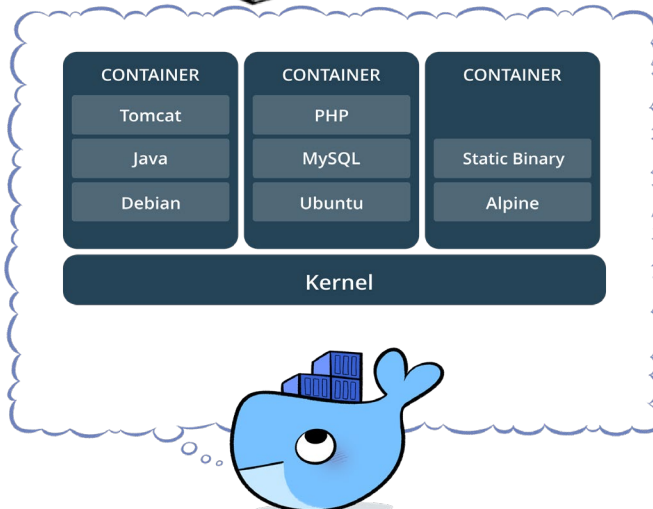
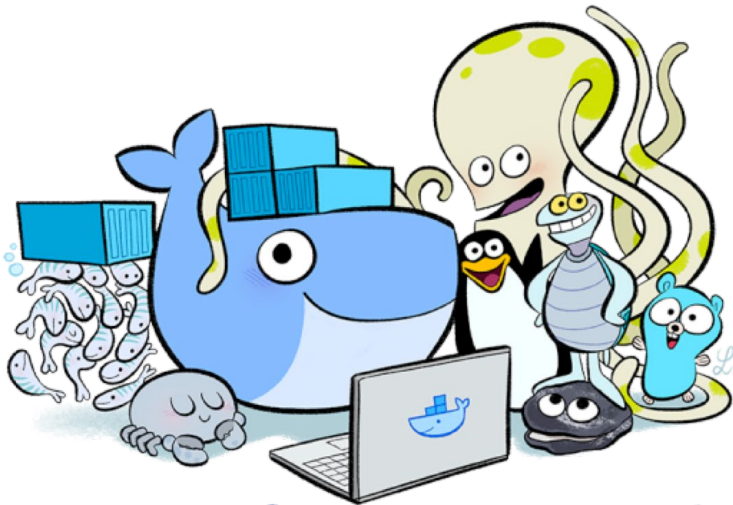
- **When extending your datacenter to the cloud.** An organization can extend the capabilities of its own on-premises network by creating a virtual network in Azure and adding VMs to that virtual network.
- **During disaster recovery.** As with running certain types of applications in the cloud and extending an on-premises network to the cloud, you can get significant cost savings by using an IaaS-based approach to disaster recovery. If a primary datacenter fails, you can create VMs running on the cloud to run your critical applications and then shut them down when the primary datacenter becomes operational again.

Containerization Fundamentals



- Containers are **lightweight, virtualized application environments**. They're designed to be quickly created, scaled out, and stopped dynamically. You can **run multiple instances of a containerized application on a single host machine**.
- Container Instances and orchestration layers, such as **Kubernetes**, are compute resources that you can use to deploy and manage containers.

Containerization Fundamentals



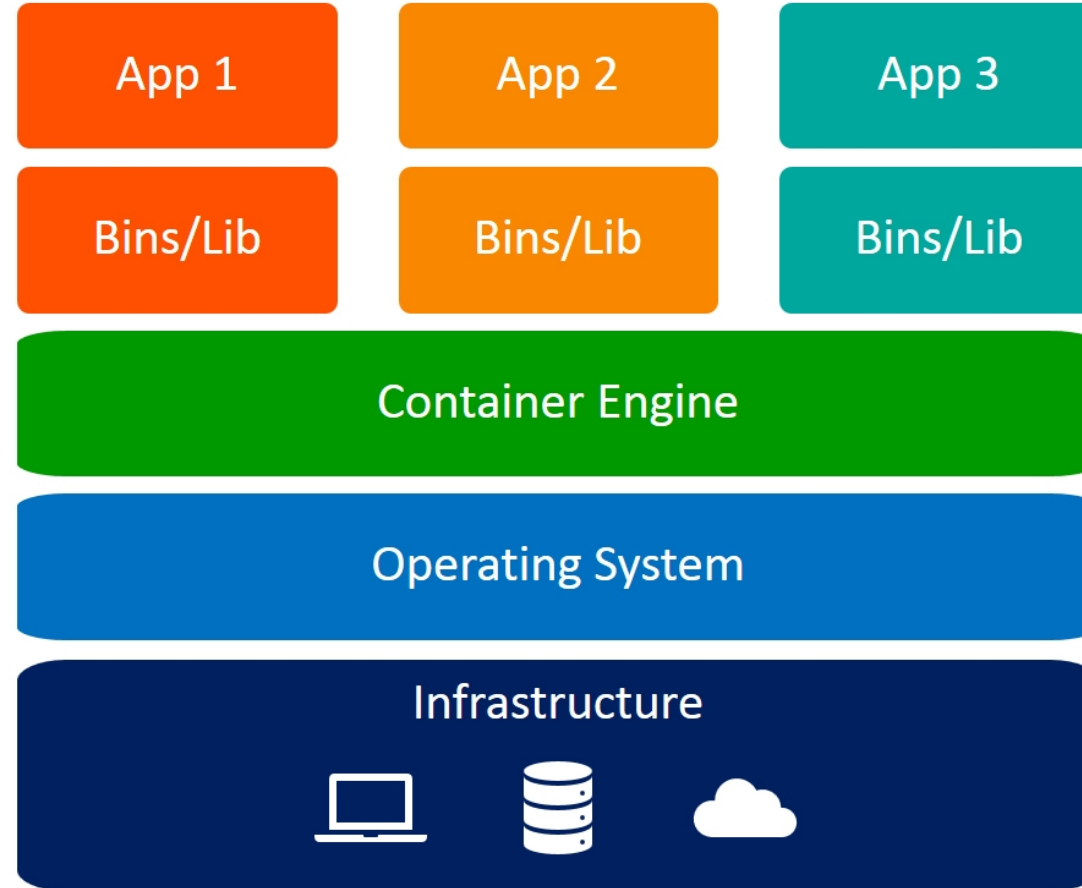
Containers



- While virtual machines are an excellent way to reduce costs versus the investments that are necessary for physical hardware, VMs are still limited to a single operating system per virtual machine.
- If you want to run multiple instances of an application on a single host machine, while maintaining strong segregation, containers are an excellent choice.

Containerization Fundamentals

Feature	Virtual Machines (VMs)	Containers
Definition	Virtualized instance of an operating system with its own kernel	Lightweight instance that shares the host OS kernel
Isolation Level	Full isolation, including separate OS and kernel	Process-level isolation; shares the host OS kernel
Startup Time	Slower, usually minutes	Fast, usually seconds
Resource Efficiency	Higher overhead due to OS duplication	Lower overhead; only application and dependencies
Size	Large, often several gigabytes due to OS	Lightweight, typically megabytes
Boot Process	Boots a full OS instance	Starts the application directly
Security	Strong isolation as each VM runs in a separate OS environment	Less isolation; relies on OS-level security
Portability	Portable across hypervisors and infrastructure	Highly portable across environments and platforms
Management and Orchestration	Managed by hypervisors (e.g., VMware, Hyper-V)	Managed by container orchestrators (e.g., Kubernetes, Docker Swarm)
Use Case	Ideal for running multiple OS types on the same hardware	Ideal for microservices, lightweight applications
Resource Sharing	Less efficient as each VM requires a full OS	More efficient; containers share host OS resources
Dependency Management	Complex; dependencies managed per VM	Simplified; dependencies packaged with application
Common Tools	VMware, Hyper-V, VirtualBox	Docker, Kubernetes, Docker Swarm



Containers

Containers

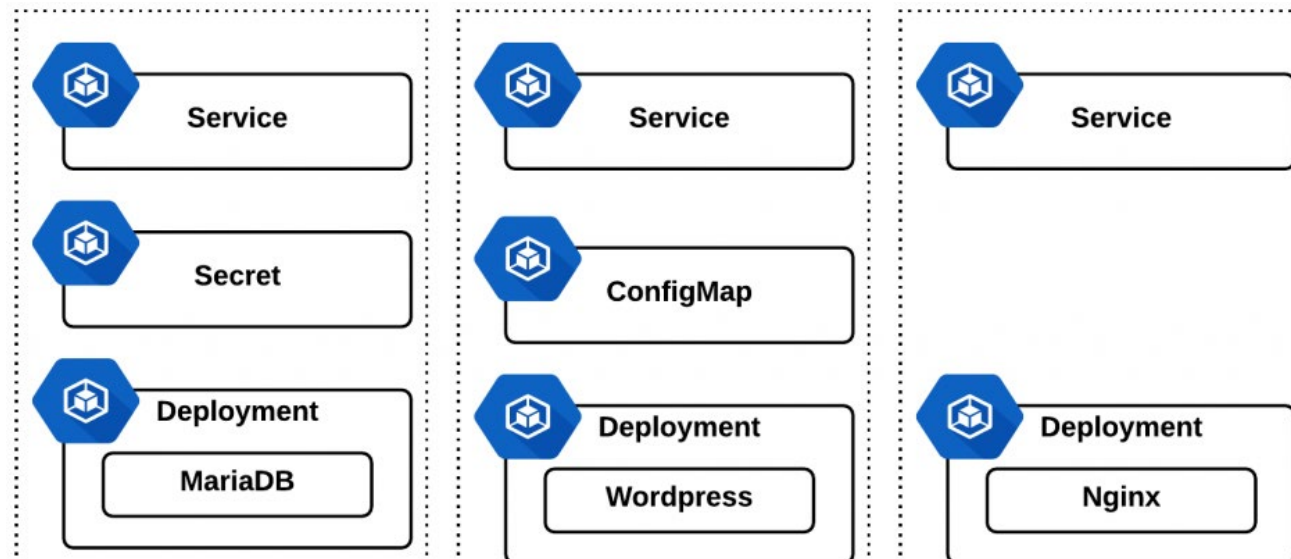


Containers

- Containers are managed through a management layer, which can start, stop, and scale out application instances as needed.
- *Standalone Container Instances* - Container Instances offers the fastest and simplest way to run a container in the cloud without having to manage any virtual machines or adopt any additional services. *It's a platform as a service (PaaS) offering that allows you to upload your containers, which it runs for you.*
- *Kubernetes* - The task of automating, managing, and interacting with a large number of containers is known as *orchestration*. Kubernetes is a complete orchestration service for containers with distributed architectures and large volumes of containers.

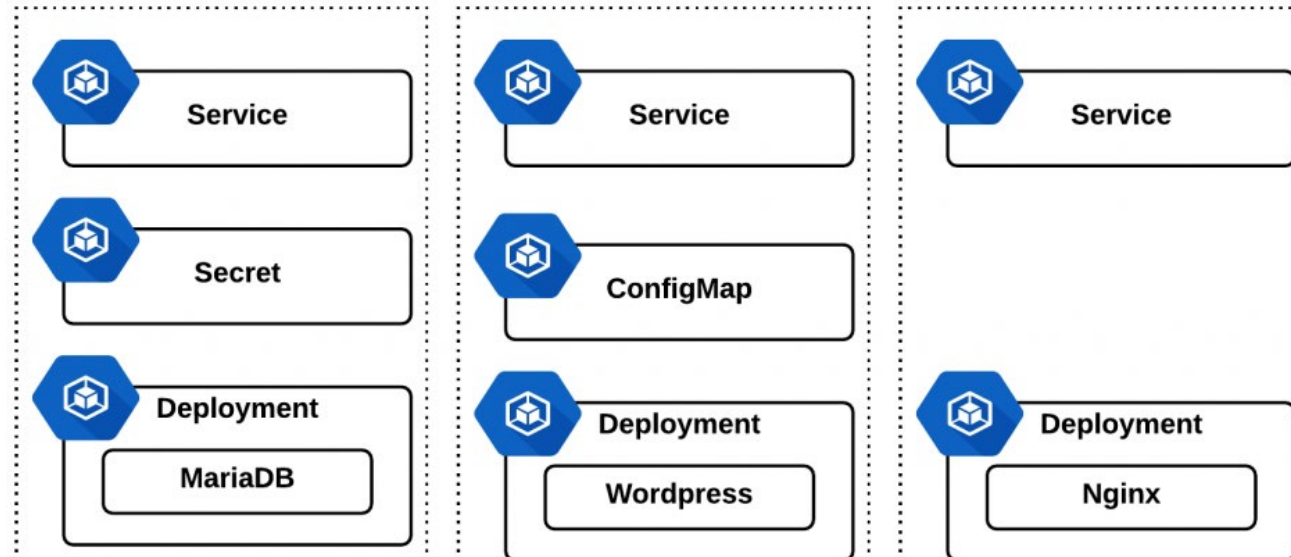
Containers are often **used to create solutions by using a *microservice architecture***. This architecture is where you break solutions into smaller, independent pieces.

For example, you might split a website into a container hosting your front end, another hosting your back end, and a third for storage. This split allows you to separate portions of your app into logical sections that can be maintained, scaled, or updated independently.



Imagine your website back-end has reached capacity but the front end and storage aren't being stressed. You could:

- Scale the back end separately to improve performance.
- Decide to use a different storage service.
- Replace the storage container without affecting the rest of the application.



Container Orchestration & PaaS

The need for orchestration (scaling, networking, healing)

Managing Containers at Scale: Beyond Single Instances

What happens when you have multiple containers?

Solution: Orchestration

- **Automatic Scaling:** Scale containers based on CPU, memory, or custom metrics
- **Service Discovery:** Containers automatically find and communicate with each other
- **Self-Healing:** Automatically restart failed containers and replace unhealthy instances
- **Rolling Updates:** Deploy new versions without downtime
- **Load Balancing:** Distribute traffic across multiple container instances

Essentially, we have a lot of data to store and process in order to keep hyperscale services running. Previous approaches cannot scale anymore.

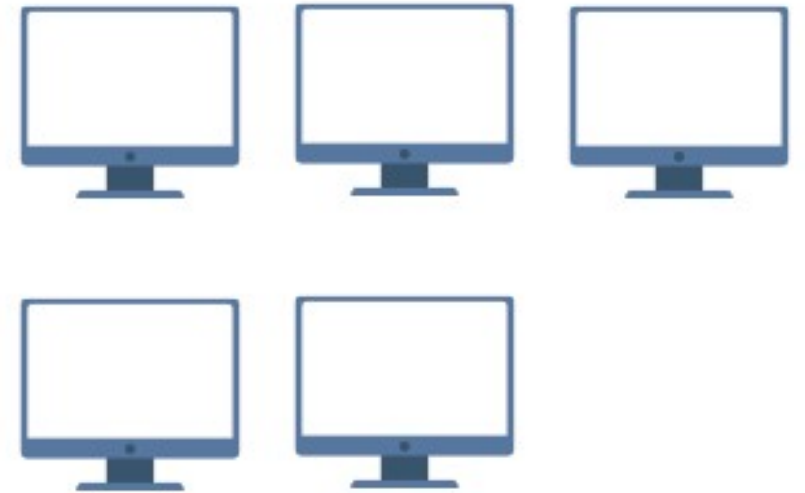


The cloud is changing the way applications are designed:

- Instead of monoliths, applications are decomposed into smaller, decentralized services.
- These services communicate through APIs or by using asynchronous messaging or eventing.
- Applications **scale horizontally**, adding new instances as demand requires.

Horizontal Scaling

Add more instances



Azure Container Instances

Launch containers with hypervisor isolation.

Get started with Azure

Container Orchestration & PaaS

[Overview](#)

[Capabilities](#)

[Security](#)

[Pricing](#)

[Resources](#)

[Next steps](#)

What can you build with Azure Container Instances?

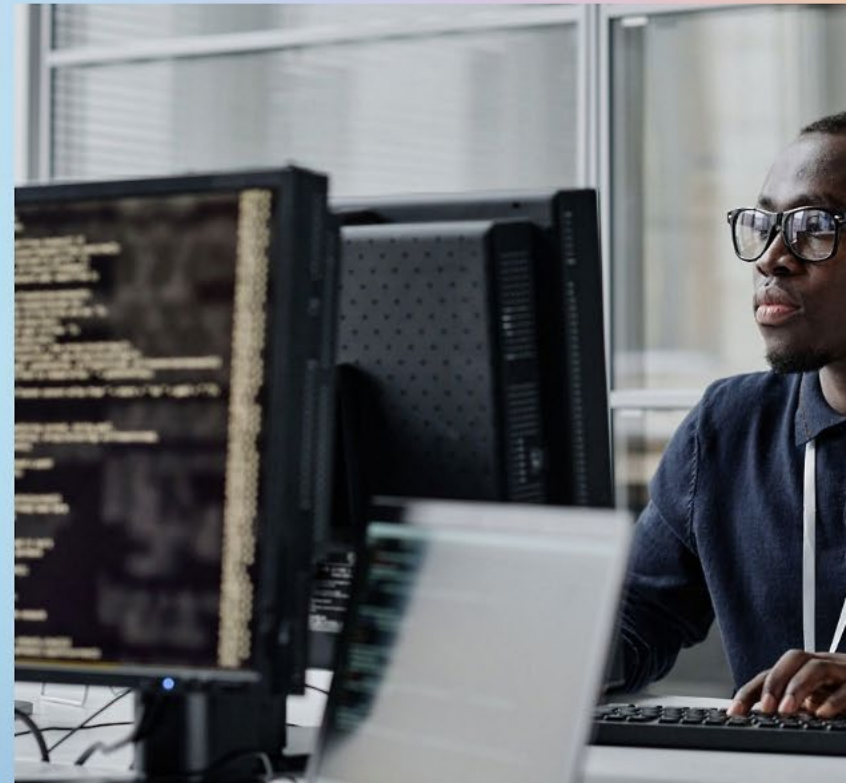
Develop apps fast without managing virtual machines or learning new tools—it's just your application, in a container, running in the cloud.

Elastic bursting with Azure Kubernetes Service (AKS) ^

Container Instances provides fast, isolated compute to meet traffic that comes in spikes, without the need to manage servers. For example, AKS uses the Virtual Kubelet to provision pods inside Container Instances that start in seconds. AKS runs with just enough capacity for your average workload. As you run out of capacity in your AKS cluster, scale out additional pods in Container Instances without additional servers to manage.

Event-driven applications with Azure Logic Apps v

Data processing jobs v



Azure Container Instances

Launch containers with hypervisor isolation.

Get started with Azure

Overview

Capabilities

Security

Pricing

Resources

Next steps

Container Orchestration & PaaS

Container Instances pricing

- No upfront cost
- No termination fees
- Per-second billing

[Learn more about Container Instances pricing](#)

CAPABILITIES

Key Features



Run containers without managing servers

Running your workloads in Container Instances empowers you to focus on designing and building your applications instead of managing the infrastructure that runs them.



Increase agility with containers on demand

Deploy containers to the cloud with unprecedented simplicity and speed—with a single command. Use Container Instances to provision additional compute for demanding workloads whenever you need to. For example, with the Virtual Kubelet, use Container Instances to elastically burst from your Azure Kubernetes Service (AKS) cluster when traffic comes in spikes.



Secure applications with hypervisor isolation

Gain the security of virtual machines for your container workloads, while preserving the efficiency of lightweight containers. Container Instances provides hypervisor isolation for each container group so containers run in isolation without sharing a kernel.

Container Orchestration & PaaS



Azure

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Microsoft has been named a Leader in the 2025 Gartner® Magic Quadrant for Cloud Managed Kubernetes Services

Azure Kubernetes Service (AKS)

Deploy and scale containers on managed Kubernetes.



Simplify Kubernetes operations, build cloud-native apps, and innovate with AI and open-source technology.

Get started with Azure

Public preview



AKS Automatic

For app modernization and intelligent apps
Easiest managed Kubernetes experience

Production
ready by default

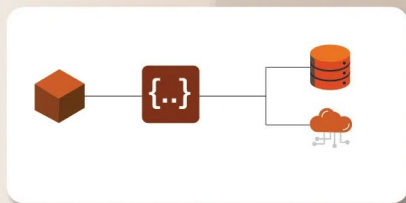
Built in best practices
and safeguards

Code to Kubernetes
in minutes

Platform & Serverless (PaaS/FaaS)

Distributed App Services

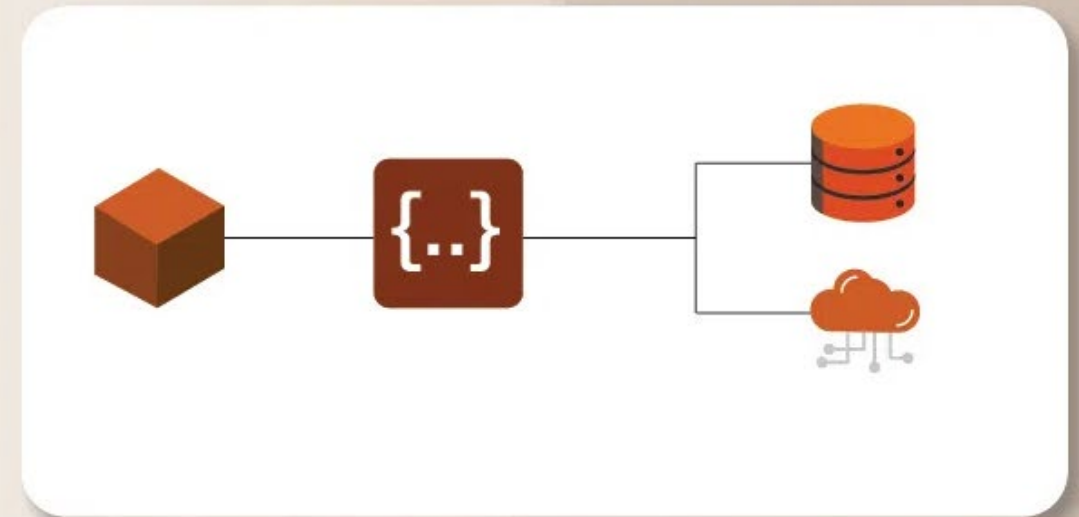
Function as a Service
(FaaS)





Serverless architecture, also known as serverless computing or Function as a Service (FaaS), is a cloud computing model where cloud providers dynamically manage the allocation and provisioning of servers, enabling programmers to concentrate only on creating and implementing functions as code.

Function as a Service (FaaS)



Home / Services / App Service

App Service

Azure App Service lets you create apps faster with a one-of-a-kind cloud service to quickly and easily create enterprise-ready web and mobile apps for any platform or device and deploy them on a scalable and reliable cloud infrastructure.

Migrate to Azure App Service

Build web apps >



Product overview Features Security Pricing Getting started Documentation Customer stories Updates More

Enabling productivity and innovation for more than 2 million apps and sites on Azure

Quickly build, deploy, and scale web apps and APIs on your terms. Work with [.NET](#), [.NET Core](#), [Node.js](#), [Java](#), [Python](#), or PHP in containers, or running on Windows or Linux. Meet rigorous, enterprise-grade performance, and security and compliance requirements with a trusted, fully managed service that handles more than 60 billion requests per day.

Chat with Sales

App Services enable you to build and host **web apps**, **background jobs**, **mobile back-ends**, and RESTful APIs in the **programming language of your choice** without managing infrastructure.

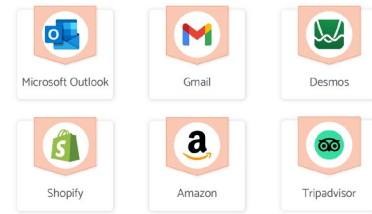
It offers automatic scaling and high availability.

This platform as a service (**PaaS**) environment allows you to focus on the website and API logic while Azure handles the infrastructure to run and scale your web applications.

With Azure App Service, you can host most common app service styles such as:

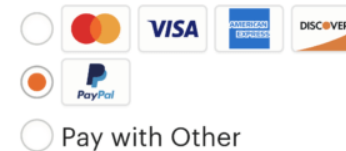
Web apps

Web application you may use daily



API apps

How you'll pay



WebJobs

WebJobs is a feature of **Azure** App Service that enables you to run a program or script in the same instance as a web app, API app, or mobile app.

Mobile apps



Web apps

App Service includes full support for hosting web apps by using **ASP.NET, ASP.NET Core, Java, Ruby, Node.js, PHP, or Python**. You can choose either Windows or Linux as the host operating system.

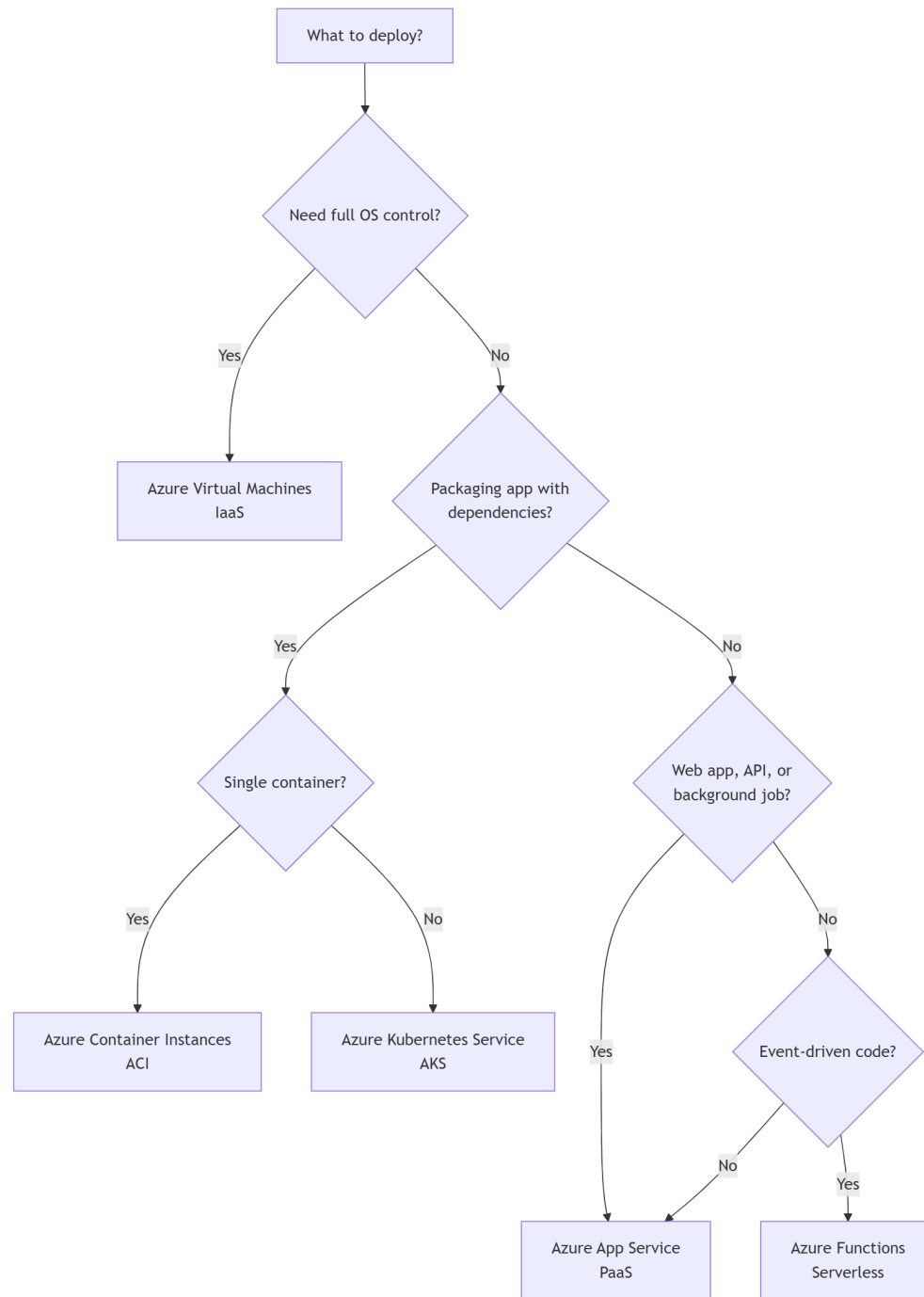
API apps

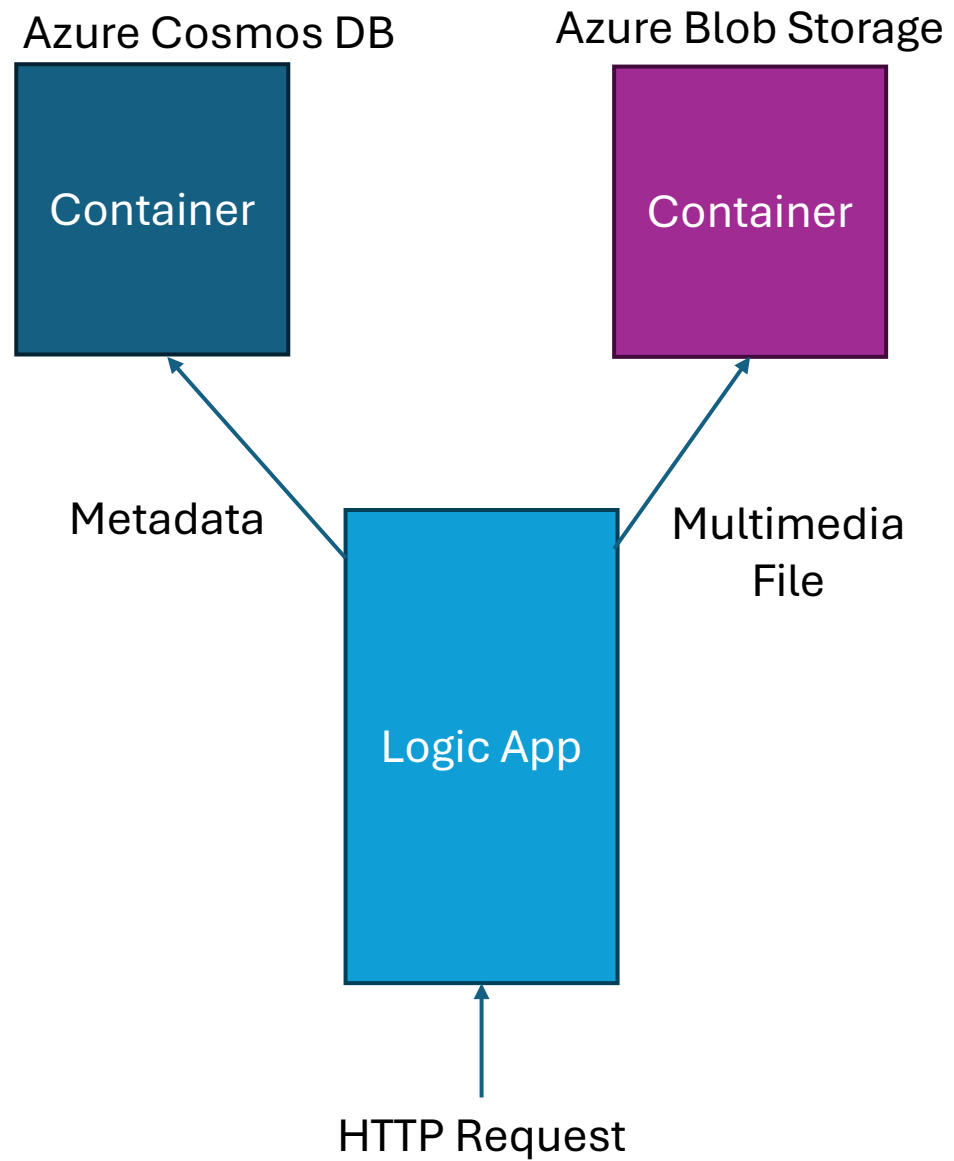
Much like hosting a website, you can build **REST-based web APIs** by using your choice of language and framework. You get full Swagger support and the ability to package and publish your API in Azure Marketplace. The produced apps can be consumed from any HTTP- or HTTPS-based client.

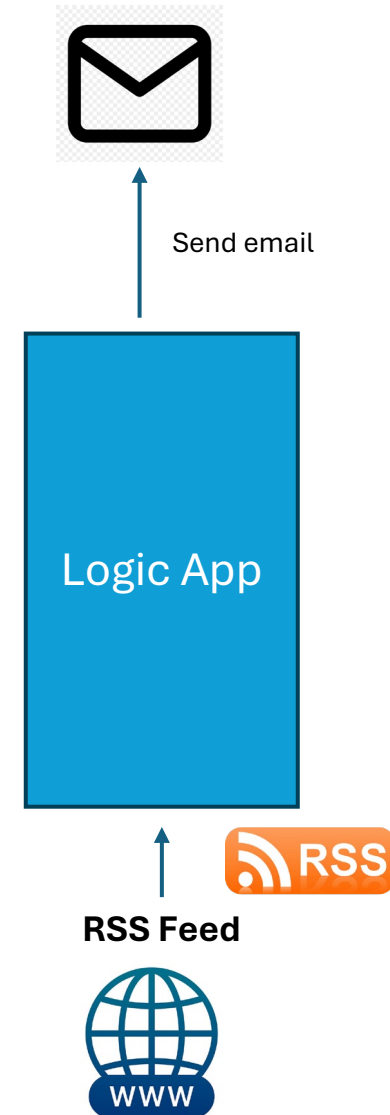
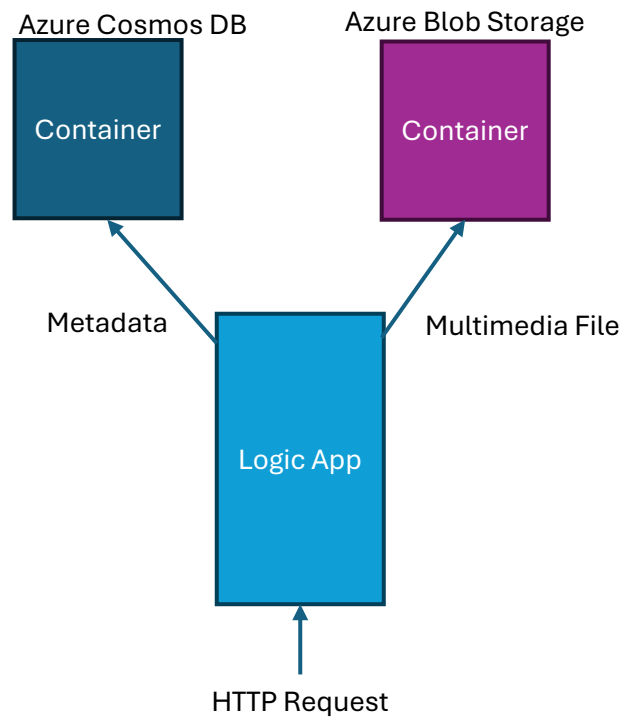
WebJobs

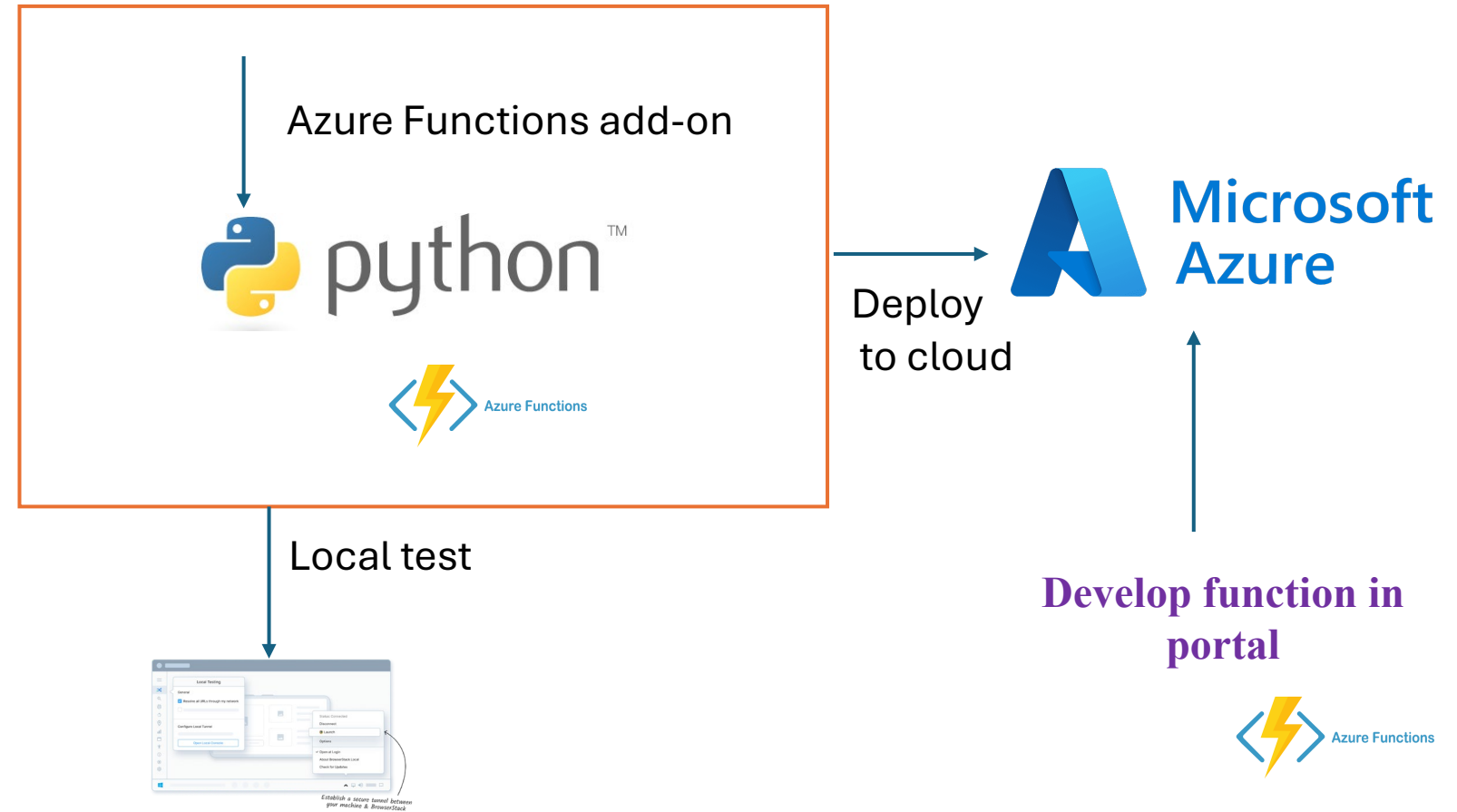
You can use the WebJobs feature to run a program (**.exe, Java, PHP, Python, or Node.js**) or script (**.cmd, .bat, PowerShell, or Bash**) in the same context as a web app, API app, or mobile app. They can be scheduled or run by a trigger. **WebJobs are often used to run background tasks as part of your application logic.**

Comparison & Decision Guide









Any Questions?

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